MIL-L-21652A(ASG)
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Superseding
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MILITARY SPECIFICATION

LIGHT, BEACON, ANTICOLLISION, AIRCRAFT, GENERAL SPECIFICATION FOR

This specification has been approved by the Department of the Air Force and by the Bureau of Naval Weapons.

1. SCOPE

1.1 Scope.- This specification covers flashing, aircraft anticollision lights.

1.2 Classification.- The lights shall be classified as follows (see 6.2):

Type I	Tandem oscillating - 115-volt, 400-cycle alternating current (ac) (MS21972)
Type II	Single rotating - 28-volt direct current (dc) (MS25277)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys

FSC 6220

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MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-L-6880	Lubrication of Aircraft, General Specification for
MIL-S-7502	Sealing Compound, Integral Fuel Tanks and Fuel Cell Cavities, High Adhesion, Accelerator Required
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-M-7969	Motors, Alternating Current, 400-Cycle, 115/200-Volt System, Aircraft, Class A and Class B, General Specification for
MIL-C-7989	Covers, Light-Transmitting, for Aeronautical Lights, General Specification for
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-M-8609	Motors, Direct-Current, 28-Volt System, Aircraft Class A and B, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS33586	Metals, Definition of Dissimilar

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The lights furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable Qualified Products List.

3.2 Data.- Unless otherwise specified in the contract or order, no data (other than reports and drawings accompanying qualification samples) are required by this specification or any of the documents referenced in section 2 (see 6.2).

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3.3 Materials.- Materials shall conform to applicable specifications as specified herein and on the application standards. Materials which are not covered by applicable specifications or standards or which are not specifically described herein shall be of the best quality and suitable for the purpose intended.

3.3.1 Metals.- Wherever practicable, all metals used in the manufacture of the lights shall be corrosion resistant unless plated or treated to resist corrosion. The use of dissimilar metals shall be avoided whenever practicable or used in accordance with MS33586.

3.3.2 Protective treatment.- When materials are used in the construction of the lights that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.3 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.3.3.1 Use of commercial parts.- Bolts, nuts, washers, cotter pins, lock rings and similar fastening devices used for assembly, may be selected from commercial sources, provided Military or other standard parts are not specifically called out by this specification or associated standards. If such commercial parts are used, they shall possess suitable properties and shall be replaceable by Military Standard (MS) parts without alteration, and provided the corresponding Military Standard part numbers are referenced in the parts list, and, if practicable, on the contractor's drawings.

3.4 Design and construction.- The lights shall be designed in accordance with the applicable MS. Optional designs may be used only if approved by the procuring activity.

3.4.1 The lights shall be so designed and constructed that no parts will work loose in service. They shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The lights shall be so constructed that adjustments and repairs can easily be made by the personnel of operating units and overhaul bases without the use of special tools.

3.4.2 Shielding.- The light shall be designed to accommodate a shield, when required, between the lamp and the cover glass to prevent the projection of the light beam into the cockpit or other crew enclosures (see 6.2).

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3.4.3 Glass cover.- A glass cover conforming to class B of MIL-C-7989 shall be mounted to provide an unobstructed transmission of light beams.

3.4.3.1 Color.- The color of all light emitted shall be aviation red conforming to MIL-C-25050.

3.4.4 Drainage.- The lights shall be designed so that sufficient drainage of water vapor condensate will be provided through drain holes or other means when mounted in the upright or inverted position.

3.4.5 Dimensions.- The dimensions of the lights shall conform to the applicable MS.

3.4.6 Installation.- The lights shall be designed for installation from the outside of the fuselage.

3.4.7 Connectors.- Electrical connectors shall be as shown on the applicable MS and shall be connected as follows:

Type I light	Type II light
Terminal A - Circuit	Terminal A - Lamp circuit
Terminal B - Common ground	Terminal B - Motor circuit
	Terminal C - Common ground

3.4.8 Housing.- The mechanisms and electrical wiring shall be suitably enclosed or protected to prevent the entrance of foreign matter.

3.4.9 Mounting.- The general design and mounting screw locations shall conform to the applicable MS.

3.4.9.1 Sealant.- A sealant conforming to MIL-S-7502 shall be used to seal the mounting flange to the housing in accordance with the applicable MS.

3.4.9.2 Gasket.- A gasket shall be provided to permit a seal between the mounting flange and the mating surface of the aircraft.

3.4.10 Lamps.- The lamps shall be as specified on the applicable MS.

3.4.11 Motors.- All dc motors shall be in accordance with MIL-M-8609. All ac motors shall be in accordance with MIL-M-7969.

3.4.12 Motor circuit.- The lights shall be designed to operate satisfactorily on the supply voltages specified in tables I and II of MIL-STD-704.

3.4.13 Voltage.- Voltage for type I light shall be 115 volts, 400 cycles ac. Type II light voltage shall be 28 volts dc.

3.5 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-70327.

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3.6 Finishes and protective coatings.-

3.6.1 Anodizing.- All aluminum and aluminum alloy parts shall be anodized in accordance with MIL-A-8625. Aluminum and aluminum alloy parts, where the finish is for corrosion resistance or paint base, shall be protected by a suitable chemical film process in accordance with MIL-C-5541.

3.6.2 Cadmium plating.- All steel parts, except lubricated internal working parts and those fabricated from corrosion resistant steel, shall be cadmium plated in accordance with QQ-P-416.

3.6.3 Paint.- All exposed parts of the anticollision light, except bearing surfaces and the lamp, shall be finished with one coat of zinc-chromate primer conforming to MIL-P-8585 before assembly and two coats zinc-chromate primer pigmented gray after assembly.

3.7 Screw threads.- Screw threads shall be in accordance with MIL-S-7742.

3.8 Weight.- The weight of one light, including the lamps, shall not exceed the weight shown on the applicable MS.

3.9 Lubricant.- Lubrication, where required, shall be in accordance with MIL-L-6880.

3.10 Performance.- The lights shall operate without failure in accordance with the requirements specified in table I and 3.10.1, when tested as specified in the applicable paragraph of section 4:

TABLE I. Performance requirements

Requirement	Test paragraph
Operation at room temperature (approx. 77° F)	4.6.2
Flash rate: 90 ±9 flashes per minute	4.6.3
Radio interference between 150 kilocycles (kc) and 399.9 megacycles (mc)	4.6.4
Life: 500 hours' operation	4.6.5
Low temperature: Procedure I of MIL-E-5272	4.6.6(a)
High temperature: Procedure I of MIL-E-5272	4.6.6(b)
Salt spray: Procedure II of MIL-E-5272 for 50 hours	4.6.6(c)
Humidity: Procedure I of MIL-E-5272	4.6.6(d)
Sand and dust: Procedure I of MIL-E-5272	4.6.6(e)
Vibration: Procedure XII of MIL-E-5272	4.6.6(f)
Altitude: Procedure VI Condition C of MIL-E-5272	4.6.6(g)
Fungus: Procedure I of MIL-E-5272	4.6.6(h)

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3.10.1 Light intensity.- The minimum light intensity for the light in all vertical planes measured with the red color filter in place, and expressed in terms of effective candlepower as determined by the formula below shall equal or exceed the values specified in table II when tested as specified in 4.6.7.

$$I_e = \frac{\int_{t_1}^{t_2} I(t)\, dt}{0.2 + (t_2 - t_1)}$$

Where I_e = Effective candlepower
$I(t)$ = Instantaneous candlepower as a function of time
$(t_2 - t_1)$ = Flash time interval in seconds

TABLE II. Light intensity

Angle above and below mounting plane of light in degrees	Minimum effective candlepower, red light required
0 to 5	100
5 to 10	60
10 to 20	20
20 to 30	10

3.11 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12 Workmanship.- The light, including all parts and accessories, shall be constructed and finished to produce an item free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, finish, alignment of parts, tightness of screw assemblies, and freedom of parts from burrs and sharp edges.

3.12.1 Cleaning.- The lights shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign material shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections.- The examinations and tests of lights shall be classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection.- Qualification inspection shall consist of all examinations and tests of this specification specified under 4.6.

4.3.1 Sampling.- Samples shall consist of two lights upon which qualification is desired. One of these lights shall have been tested by the manufacturer in accordance with this specification prior to being forwarded to the activity responsible for qualification. These lights shall be accompanied by one complete set of detail and assembly drawings and a complete test report showing results of the manufacturer's tests. The samples shall be plainly identified by securely attached durable tags marked with the following information and forwarded to the activity responsible for qualification, designated in the letter of authorization from that activity (see 6.3).

Sample for qualification inspection
LIGHT, BEACON, ANTICOLLISION, AIRCRAFT
Manufacturer's part number
Name of manufacturer
Submitted by (name) (date) for qualification test in accordance with the requirements of MIL-L-21652A(ASG) under authorization (reference letter of authorization).

4.3.2 Lamps.- When the tests are conducted, the sample lights shall be equipped with the lamps for which they were designed.

4.3.3 Manufacturer's drawings.- Manufacturer's drawings submitted with the qualification inspection samples shall show all parts in their normal assembled position and shall specify part numbers of all parts and subassemblies. The drawings shall be in accordance with MIL-D-70327. The following shall be furnished on, or together with, the assembly drawings:

(a) Overall dimensions.
(b) Materials and construction, treatment and finish.
(c) Motor electrical characteristics.
(d) Maximum permissible temperature rise of motor.

4.3.4 Manufacturer's test report.- The test report submitted with the qualification inspection samples shall include reports of all tests indicating compliance with all requirements of this specification, referring specifically to paragraph numbers.

4.4 Quality conformance inspection.- Quality conformance inspection shall consist of the examinations and tests listed in table III.

4.4.1 Sampling.- For each group of tests, a random sample of the size specified shall be selected from an inspection lot.

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4.4.1.1 Group I tests.- Group I tests are for minor defects. Sampling for group I tests shall be in accordance with MIL-STD-105, using an Acceptable Quality Level (AQL) of 6.5 percent defective.

4.4.1.2 Group II tests.- Group II tests are for major defects. Sampling for group II tests shall be in accordance with MIL-STD-105, using an AQL of 1.0 percent defective.

4.4.2 Samples subjected to quality conformance tests, if approved, may be considered as part of, and delivered under contract.

TABLE III. Quality conformance tests

Group	Test	Requirement	Test paragraph
I	Examination of product	—	4.6.1
II	Operation	Table I	4.6.2
	Flash rate	Table I	4.6.3

4.5 Test conditions.-

4.5.1 Standard atmospheric conditions.- When the pressure and temperature existing at the time of the test are not specified definitely, the test shall be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 77° F). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference.

4.6 Test methods.- The applicable tests shall be performed as follows:

4.6.1 Examination of product.- The light shall be carefully examined to determine conformance to all the requirements of this specification for which no specific tests are described and to determine conformance to manufacturer's approved material, fabrication, specifications, and drawings.

4.6.2 Operation.- The light shall be operated at room temperature (approximately 77° F) from the power source specified on the MS. There shall be no evidence of malfunction or overheating of the motor.

4.6.3 Flash rate.- The light shall be tested with the voltage specified on the MS to determine that it conforms to the flash rate specified in table I.

4.6.4 Radio interference.- The light shall pass the radio interference requirements test specified in the MIL-I-6181, except that the test shall be conducted only between 150 kc. and 399.9 mc.

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4.6.5 Life.- The light shall be mounted under simulated conditions of actual installation, except provisions for inverting the light without interrupting operation shall be provided. The light shall be operated from a source as specified on the applicable MS. The mechanism shall be operated 250 hours in the upright position and 250 hours in the inverted position. There shall be no excessive wear, failure, or malfunction of any part, except lamp filament. Room temperature blast cooling air may be provided for 23 out of each 24 hours of test.

4.6.6 Environmental tests.- The environmental tests shall be in accordance with specified procedures of MIL-E-5272 and as specified herein:

(a) Low temperature - Procedure I: The light shall be examined and then operated to determine proper starting, operation, and extent of damage.

(b) High temperature - Procedure I: The light shall be examined and then operated to determine proper starting, operation, and extent of damage.

(c) Salt spray: Procedure II, except that the light shall be subjected to the salt spray test for 50 hours. If the lamp burns out, it shall be replaced. There shall be no evidence of excessive corrosion or failure to operate of any part of the light.

(d) Humidity - Procedure I: The light shall be visually inspected for evidence of corrosion. It shall operate satisfactorily at the conclusion of the test.

(e) Sand and dust - Procedure I: There shall be no damage to any of the moving parts of the light.

(f) Vibration - Procedure XII: There shall be no failure of any part of the light, exclusive of the lamp filament.

(g) Altitude - Procedure VI, Condition C: There shall be no failure of any part of the light.

(h) Fungus - Procedure I: There shall be no deterioration or corrosion of any internal or external components which could in any manner prevent the light from meeting operational, maintenance, or servicing requirements during service life.

4.6.7 Light intensity.- The light shall be tested for conformance to 3.10.1.

4.7 Packaging, packing, and marking.- Preparation for delivery shall be examined for conformance to section 5.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- Each light assembly shall be prepared for delivery in accordance with MIL-P-7936. When level A preservation and packaging requirements are specified, Method IA without the use of a preservative compound shall apply.

5.2 Marking of shipments.- In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The lights are electrically operated anticollision lights intended for use as follows:

Type I - for use on high speed aircraft.
Type II - for use on slower speed aircraft.

6.2 Ordering data.- Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Applicable MS part number (see 3.4) and type (see 1.2).
(c) Quantity of lights desired.
(d) Data requirements (see 3.2).
(e) Applicable levels of packaging and packing (see 5.1).
(f) Whether lamps should be furnished with the lights (see 3.4.10).
(g) Whether shields should be furnished with the lights (see 3.4.2).

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Navy Department, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity.

6.4 It is understood that the lights offered under contract represent the identical quality of materials and workmanship to be found in the qualification samples and that no substitution or deviation from these samples will be permitted unless change is approved in advance by the contracting officer.

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